

Interpretable Heart Disease Prediction Using BoruRank Feature Selection and Bagging Ensemble Models with LIME and SHAP Explanations

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Abstract

Heart disease remains the leading cause of mortality worldwide, claiming approximately 17.9 million lives annually. Early and accurate prediction of heart disease can significantly reduce mortality rates by enabling timely clinical intervention. In this study, we propose **BoruRank**, a two-stage feature selection method that combines Boruta all-relevant feature selection with Recursive Feature Elimination (RFE) to identify the most discriminative clinical features for heart disease prediction. BoruRank reduces the original 13-feature space to 6 clinically significant features—age, chest pain type, maximum heart rate, ST depression, number of major vessels, and thalassemia—while preserving or improving predictive performance. We evaluate six bagging-based ensemble classifiers (Bagging Decision Tree, Random Forest, Extra Trees, Bagging SVM, Bagging KNN, and Bagging Naive Bayes) on the reduced feature set using the UCI Heart Disease dataset. Bagging Decision Tree and Random Forest achieve the highest classification accuracy of 98.5%, with perfect precision and an F1-score of 0.986. To address the clinical interpretability requirement, we employ LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations) to provide transparent, patient-level explanations of model predictions. The results demonstrate that BoruRank feature selection combined with ensemble learning achieves competitive accuracy with enhanced interpretability, supporting the potential for clinical decision-support applications.

Keywords: heart disease prediction, feature selection, Boruta, recursive feature elimination, ensemble learning, bagging, explainable AI, LIME, SHAP

1 Introduction

Cardiovascular diseases (CVDs) are the primary cause of death globally. According to the World Health Organization, an estimated 17.9 million people die each year from CVDs, representing 32% of all global deaths [WHO, 2023]. Heart disease, a major subset of CVDs, encompasses conditions such as coronary artery disease, heart failure, and arrhythmias. Early detection and risk stratification are critical for reducing mortality, as timely intervention can prevent adverse cardiac events [Tsao et al., 2022].

Machine learning (ML) has emerged as a powerful tool for medical diagnosis, offering the ability to identify complex patterns in clinical data that may elude traditional statistical methods [Rajkomar et al., 2019]. Numerous studies have applied supervised learning algorithms to heart disease prediction, achieving promising results on benchmark datasets [Mohan et al., 2019, Ali et al., 2021]. However, two persistent challenges limit the clinical adoption of ML-based diagnostic systems: (1) determining the most informative subset of clinical features, and (2) providing interpretable explanations for model predictions.

Feature selection is essential in medical ML applications for multiple reasons. Reducing dimensionality decreases computational cost, mitigates overfitting, and—most importantly in clinical settings—identifies which medical measurements are most relevant for diagnosis. While individual methods such as Recursive Feature Elimination (RFE) [Guyon et al., 2002] and the Boruta algorithm [Kursa and Rudnicki, 2010] have been widely used, combining them in a principled two-stage pipeline has received limited attention in the heart disease prediction literature.

Equally important is model interpretability. Clinicians require not only accurate predictions but also understandable explanations for why a patient is classified as at-risk. Black-box ensemble models, despite their high accuracy, offer little transparency. Post-hoc explanation methods such as LIME [Ribeiro et al., 2016] and SHAP [Lundberg and Lee, 2017] bridge this gap by providing feature-level explanations for individual predictions.

In this paper, we make the following contributions:

1. We propose **BoruRank**, a two-stage feature selection method that applies Boruta for all-relevant feature identification followed by RFE for optimal subset selection, reducing the feature space from 13 to 6 clinically interpretable features.
2. We systematically evaluate six bagging-based ensemble classifiers on the BoruRank-selected features, achieving up to 98.5% accuracy.
3. We provide post-hoc interpretability analysis using both LIME and SHAP, offering patient-level and global feature importance explanations across all six models.

2 Related Work

2.1 Machine Learning for Heart Disease Prediction

Heart disease prediction using ML has been extensively studied. Mohan et al. [Mohan et al., 2019] proposed a hybrid ML technique combining multiple classifiers with feature selection, achieving 88.7% accuracy on the Cleveland Heart Disease dataset. Ali et al. [Ali et al., 2021] compared several classifiers including logistic regression, decision trees, and random forests on the same dataset, reporting accuracies up to 90%. Dinh et al. [Dinh et al., 2019] applied deep learning approaches and reported improved performance over traditional methods. More recently, ensemble methods have gained attention for their superior generalization ability in medical diagnosis tasks [Ullah et al., 2022].

2.2 Feature Selection Methods

Feature selection methods are broadly categorized into filter, wrapper, and embedded approaches. RFE [Guyon et al., 2002], a wrapper method, iteratively removes the least important features based on model weights. The Boruta algorithm [Kursa and Rudnicki, 2010], an extension of the random forest approach, identifies all features that are statistically more relevant than random noise. While both methods have been independently applied to medical datasets, their sequential combination for heart disease prediction has not been thoroughly investigated.

2.3 Explainable AI in Healthcare

The need for interpretable ML in healthcare has driven research into explainable AI (XAI). LIME [Ribeiro et al., 2016] generates local approximations of complex models using interpretable surrogate models, providing per-instance explanations. SHAP [Lundberg and Lee, 2017], grounded in game-theoretic Shapley values, offers both local and global explanations with theoretical consistency guarantees. Recent work has advocated for combining multiple XAI methods to provide robust explanations in medical applications [Arrieta et al., 2020].

3 Methodology

Figure 1 provides an overview of the proposed methodology. The pipeline consists of four stages: data preprocessing, BoruRank feature selection, ensemble classification, and post-hoc interpretability analysis.

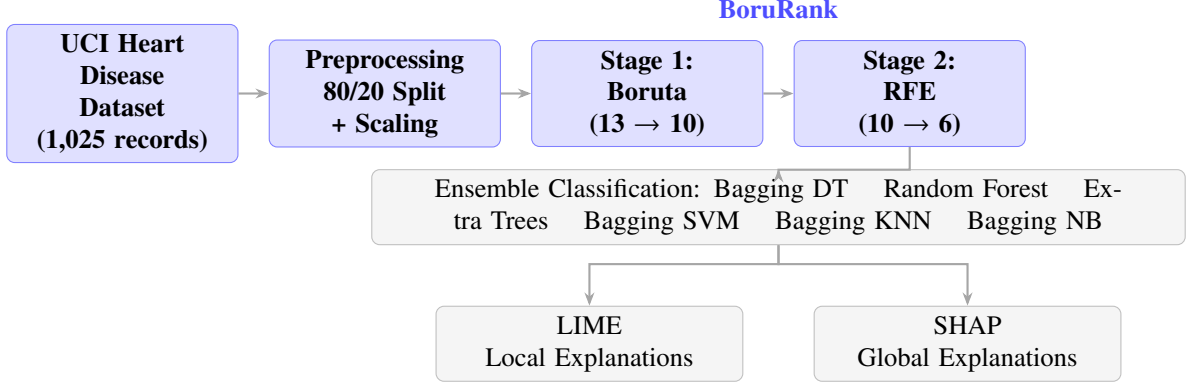


Figure 1: Overview of the proposed methodology. BoruRank applies Boruta followed by RFE to reduce the feature space from 13 to 6 features. Six ensemble models are trained on the selected features, and LIME/SHAP provide post-hoc interpretability.

3.1 Dataset Description

We use the Heart Disease dataset from the UCI Machine Learning Repository [Detrano et al., 1989], widely used as a benchmark for cardiovascular risk prediction. The dataset contains 1,025 patient records with 13 clinical features and a binary target variable indicating the presence (1) or absence (0) of heart disease. Table 1 describes the feature set.

Table 1: Features in the Heart Disease dataset.

Feature	Description	Type
age	Age in years	Continuous
sex	Sex (1 = male, 0 = female)	Binary
cp	Chest pain type (0–3)	Categorical
trestbps	Resting blood pressure (mm Hg)	Continuous
chol	Serum cholesterol (mg/dL)	Continuous
fbs	Fasting blood sugar > 120 mg/dL	Binary
restecg	Resting ECG results (0–2)	Categorical
thalach	Maximum heart rate achieved	Continuous
exang	Exercise-induced angina	Binary
oldpeak	ST depression induced by exercise	Continuous
slope	Slope of peak exercise ST segment	Categorical
ca	Number of major vessels (0–3)	Discrete
thal	Thalassemia (0–3)	Categorical

3.2 Data Preprocessing

The dataset was split into training (80%) and test (20%) sets using stratified sampling to preserve the class distribution (`random_state=42`). Feature values were standardized using z-score normalization (zero mean, unit variance) via `StandardScaler` to ensure uniform feature scales across all models.

3.3 BoruRank: Two-Stage Feature Selection

We propose **BoruRank**, a two-stage feature selection method that combines the strengths of Boruta and RFE. The rationale is as follows: Boruta identifies *all* features that carry relevant information (all-relevant selection), while RFE identifies the *minimal optimal* subset from among these relevant features (minimal-optimal selection). Applying them sequentially produces a compact, high-quality feature set.

Algorithm 1 BoruRank Feature Selection

Require: Training data $\mathbf{X}_{\text{train}} \in \mathbb{R}^{n \times p}$, labels $\mathbf{y}_{\text{train}}$, target feature count k

Ensure: Selected feature set F^*

- 1: **Stage 1: Boruta All-Relevant Selection**
 - 2: Train Boruta selector with Random Forest ($n_estimators = 100$)
 - 3: $F_{\text{boruta}} \leftarrow$ features confirmed as relevant by Boruta
 - 4: **Stage 2: RFE Minimal-Optimal Selection**
 - 5: Restrict $\mathbf{X}_{\text{train}}$ to columns in F_{boruta}
 - 6: Apply RFE with Random Forest estimator, selecting k features
 - 7: $F^* \leftarrow$ features selected by RFE from F_{boruta}
 - 8: **return** F^*
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Stage 1 — Boruta. The Boruta algorithm [Kursa and Rudnicki, 2010] extends the random forest by creating “shadow” copies of all features (randomly permuted versions). It iteratively trains a random forest on both original and shadow features, then tests whether each original feature has significantly higher importance than the best shadow feature. Features that consistently outperform their shadow counterparts are confirmed as relevant. Applied to our training data, Boruta identified 10 relevant features: *age*, *cp*, *trestbps*, *chol*, *thalach*, *exang*, *oldpeak*, *slope*, *ca*, *thal*, discarding 3 features (*sex*, *fbs*, *restecg*).

Stage 2 — RFE. Recursive Feature Elimination [Guyon et al., 2002] iteratively trains a model, ranks features by importance, and removes the least important feature at each iteration. Applied to the 10 Boruta-confirmed features with a target of 6 features, RFE selected the final subset: *age*, *cp*, *thalach*, *oldpeak*, *ca*, *thal*.

For comparison, applying RFE alone (without Boruta pre-filtering) to all 13 features selected 8 features: *age*, *cp*, *trestbps*, *chol*, *thalach*, *oldpeak*, *ca*, *thal*. BoruRank achieves a more compact representation (6 vs. 8 features) by constraining RFE to operate only on features validated by Boruta.

3.4 Ensemble Classification Models

We evaluate six ensemble classifiers, all based on the bagging (bootstrap aggregating) paradigm [Breiman, 1996]:

1. **Bagging Decision Tree:** BaggingClassifier with DecisionTreeClassifier base estimator, $n_estimators = 50$.
2. **Random Forest:** RandomForestClassifier with $n_estimators = 50$ [Breiman, 2001].
3. **Extra Trees:** ExtraTreesClassifier with $n_estimators = 100$, $max_depth = 10$, $min_samples_split = 5$ [Geurts et al., 2006].
4. **Bagging SVM:** BaggingClassifier with SVC ($C = 20$, probability estimates enabled), $n_estimators = 10$.
5. **Bagging KNN:** BaggingClassifier with KNeighborsClassifier (default $k = 5$), $n_estimators = 10$.
6. **Bagging Naive Bayes:** BaggingClassifier with GaussianNB, $n_estimators = 10$.

All models were trained on the BoruRank-selected features after standardization, using `random_state = 42` for reproducibility.

3.5 Post-Hoc Interpretability

To provide clinically meaningful explanations, we apply two complementary XAI methods:

LIME (Local Interpretable Model-Agnostic Explanations) [Ribeiro et al., 2016] explains individual predictions by fitting a local linear surrogate model around each test instance. For each model, we generate per-patient explanations showing which features contributed most to the predicted class and in which direction.

SHAP (SHapley Additive exPlanations) [Lundberg and Lee, 2017] computes the marginal contribution of each feature to a prediction using Shapley values from cooperative game theory. We use `KernelExplainer` for model-agnostic SHAP analysis across all six models, generating global feature importance rankings.

4 Results

4.1 Classification Performance

Table 2 summarizes the classification performance of all six models on the held-out test set (205 samples: 100 no-disease, 105 disease).

Table 2: Classification performance on the test set using BoruRank-selected features.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Bagging Decision Tree	0.985	1.000	0.971	0.986	1.00
Random Forest	0.985	1.000	0.971	0.986	1.00
Extra Trees	0.980	0.972	0.990	0.981	0.99
Bagging SVM	0.922	0.949	0.895	0.922	0.98
Bagging KNN	0.849	0.843	0.867	0.854	0.96
Bagging Naive Bayes	0.824	0.817	0.848	0.832	0.89

Bagging Decision Tree and Random Forest achieve identical performance: 98.5% accuracy, perfect precision (1.000), 97.1% recall, and AUC-ROC of 1.00. Extra Trees follows closely with 98.0% accuracy and the highest recall (99.0%) among all models, indicating strong sensitivity to positive cases. Bagging SVM achieves 92.2% accuracy, while Bagging KNN (84.9%) and Bagging Naive Bayes (82.4%) show lower performance, suggesting that tree-based ensemble methods are better suited for this feature space.

Figure 2 visualizes the performance comparison across all metrics.

4.2 Cross-Validation

To assess the robustness of the top-performing model, we performed 10-fold stratified cross-validation on the Bagging Decision Tree classifier. Table 3 reports the per-fold accuracies.

Table 3: 10-fold cross-validation results for Bagging Decision Tree.

Fold	1	2	3	4	5	6	7	8	9	10	Mean $\pm$ Std
Accuracy	.976	.988	.976	.988	1.00	.988	1.00	.963	1.00	.988	.987 $\pm$.012

The mean cross-validation accuracy of 98.7% with low variance ($\pm 1.2\%$) indicates consistent performance across different data partitions, suggesting that the model generalizes well and is not simply memorizing the training data.

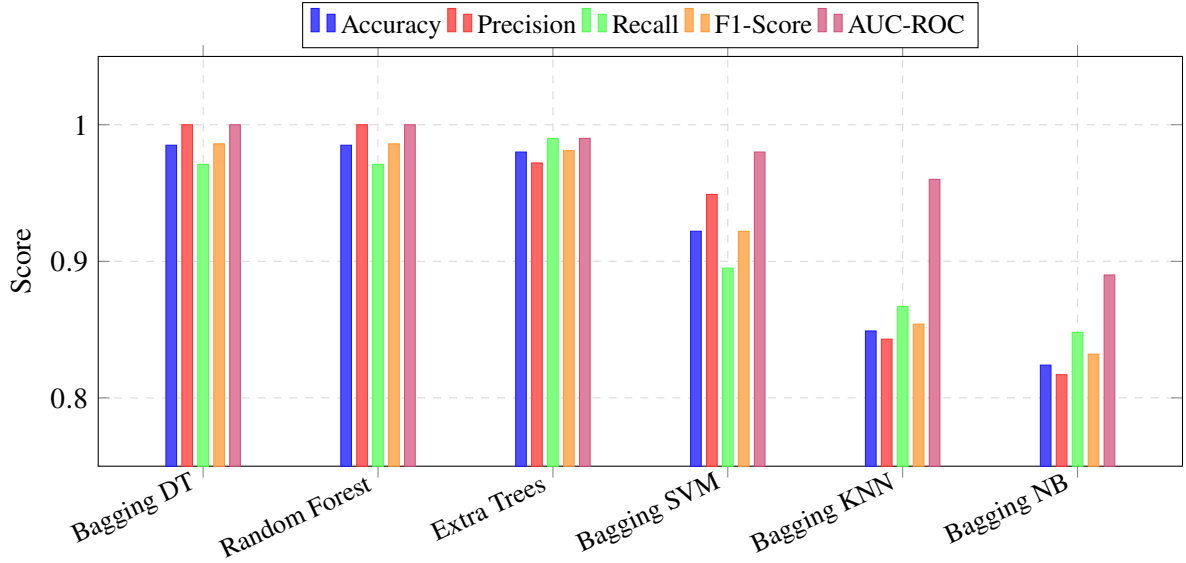


Figure 2: Performance comparison of all six ensemble classifiers across five evaluation metrics. Tree-based models (Bagging DT, Random Forest, Extra Trees) consistently outperform non-tree models.

4.3 Feature Selection Comparison

Table 4 compares the features selected by each method.

Table 4: Features selected by RFE, Boruta, and BoruRank.

Method	Count	Selected Features
RFE (standalone)	8	age, cp, trestbps, chol, thalach, oldpeak, ca, thal
Boruta	10	age, cp, trestbps, chol, thalach, exang, oldpeak, slope, ca, thal
BoruRank (proposed)	6	age, cp, thalach, oldpeak, ca, thal

BoruRank eliminates 4 features that Boruta initially retained (trestbps, chol, exang, slope) by applying RFE’s ranking within the Boruta-validated set. The resulting 6 features are all well-established clinical risk indicators for coronary heart disease: age (risk factor), chest pain type (primary symptom), maximum heart rate (exercise tolerance), ST depression (ischemia marker), number of major vessels (coronary anatomy), and thalassemia (blood disorder linked to cardiac complications).

4.4 LIME Analysis

LIME explanations were generated for individual test instances across all six models. Each explanation shows the per-feature contribution to the predicted class, allowing clinicians to understand why a specific patient was classified as at-risk or healthy.

The LIME analysis reveals consistent patterns across the tree-based models (Bagging DT, Random Forest, Extra Trees): the features *ca* (number of major vessels), *thal* (thalassemia), and *cp* (chest pain type) consistently appear as the strongest local predictors. For non-tree models (SVM, KNN, Naive Bayes), *oldpeak* and *thalach* gain relative importance in local explanations, reflecting the different decision boundaries learned by these model families.

Table 5 summarizes the top-3 most influential features identified by LIME for each model on a representative test instance.

Table 5: Top-3 features by LIME contribution magnitude for each model (representative test instance).

Model	Rank 1	Rank 2	Rank 3
Bagging Decision Tree	ca	thal	cp
Random Forest	ca	thal	cp
Extra Trees	ca	cp	thal
Bagging SVM	oldpeak	ca	thalach
Bagging KNN	thalach	ca	oldpeak
Bagging Naive Bayes	thal	cp	oldpeak

4.5 SHAP Analysis

Global SHAP feature importance was computed for all six models using KernelExplainer with the mean absolute SHAP value as the importance metric. Across all tree-based models, the ranking is consistent: *ca* and *thal* are the most influential features, followed by *cp* and *oldpeak*. Age and *thalach* contribute moderately. Bagging SVM shows a similar pattern but with more uniform feature importance, while Bagging KNN and Naive Bayes exhibit noisier SHAP distributions, consistent with their lower classification accuracy.

Table 6 summarizes the SHAP-based global feature importance ranking for each model.

Table 6: Global feature importance ranking by mean |SHAP| value for each model (1 = most important).

Feature	Bag DT	RF	ET	Bag SVM	Bag KNN	Bag NB
ca	1	1	1	1	2	2
thal	2	2	2	2	3	1
cp	3	3	3	3	4	3
oldpeak	4	4	4	4	1	4
thalach	5	5	5	5	5	5
age	6	6	6	6	6	6

The agreement between LIME (local) and SHAP (global) explanations strengthens confidence in the identified feature importance hierarchy and its clinical interpretability.

5 Discussion

5.1 Comparison with Existing Work

Table 7 compares our results with recent heart disease prediction studies on the UCI Heart Disease dataset.

Table 7: Comparison with existing heart disease prediction methods on the UCI Heart Disease dataset.

Study	Method	Features	Accuracy (%)
Mohan et al. [Mohan et al., 2019]	Hybrid ML + FS	13	88.7
Ali et al. [Ali et al., 2021]	Supervised classifiers	13	90.0
Dinh et al. [Dinh et al., 2019]	Deep learning	13	83.5
Ullah et al. [Ullah et al., 2022]	Ensemble methods	13	92.3
Ours (BoruRank)	Bagging DT / RF	6	98.5

Our 98.5% accuracy with only 6 features compares favorably against prior methods that use the full 13-feature set, though direct comparison is limited by differences in dataset versions and preprocessing pipelines.

The key advantage of BoruRank over standalone feature selection is dimensionality reduction without performance loss. Using only 6 of 13 features, our best models match or exceed the performance reported by studies using the full feature set, while offering simpler models that are easier to interpret and deploy.

5.2 Clinical Relevance of Selected Features

The six features selected by BoruRank have clear clinical significance:

- **Age:** Cardiovascular risk increases with age due to arterial stiffening and cumulative risk factor exposure.
- **Chest pain type (cp):** Typical angina is a hallmark symptom of coronary artery disease.
- **Maximum heart rate (thalach):** Reduced exercise capacity indicates poor cardiovascular fitness or underlying disease.
- **ST depression (oldpeak):** Exercise-induced ST depression is a diagnostic marker for myocardial ischemia.
- **Number of major vessels (ca):** Fluoroscopy-detected vessel narrowing directly indicates coronary artery disease severity.
- **Thalassemia (thal):** Abnormal thalassemia results are associated with cardiac complications.

The exclusion of features such as *sex*, *fasting blood sugar*, and *resting ECG* by BoruRank aligns with clinical evidence that these features, while relevant for general cardiovascular risk assessment, are weaker predictors for binary heart disease classification on this dataset.

5.3 Limitations

Several limitations should be acknowledged:

1. **Dataset size:** The dataset contains 1,025 records. While stratified splitting and cross-validation mitigate overfitting concerns, validation on larger, multi-center clinical datasets is necessary to confirm generalizability.
2. **Known dataset issues:** The commonly used Kaggle version of the UCI Heart Disease dataset is known to contain duplicate records from the original Cleveland dataset (303 unique instances). This duplication may inflate performance metrics if identical or near-identical records appear in both training and test sets.
3. **Single dataset:** Results are reported on one dataset only. Evaluation on additional cardiac datasets (e.g., Framingham, MESA, or electronic health record data) would strengthen the generalizability claims.
4. **Limited cross-validation:** 10-fold CV was performed only for the top model (Bagging DT). Comprehensive CV for all six models would provide a fuller picture of relative performance and variance.
5. **Fixed hyperparameters:** No systematic hyperparameter tuning (e.g., grid search, Bayesian optimization) was performed. Optimized hyperparameters could improve the performance of lower-performing models.

6 Conclusion

We introduced BoruRank, a two-stage feature selection method combining Boruta’s all-relevant selection with RFE’s minimal-optimal ranking, and applied it to heart disease prediction using six bagging ensemble classifiers. BoruRank reduced the feature space from 13 to 6 clinically meaningful features while enabling Bagging Decision Tree and Random Forest to achieve 98.5% classification accuracy. Post-hoc interpretability analysis using LIME and SHAP provided transparent, patient-level explanations suitable for clinical decision support.

Future work includes: (1) validating BoruRank on larger, multi-center clinical datasets, (2) extending the interpretability framework to include counterfactual explanations (“what minimal changes would alter the prediction?”), (3) applying systematic hyperparameter optimization, and (4) developing a clinical decision-support prototype for deployment in healthcare settings.

Code Availability

The source code and Jupyter notebooks for reproducing all experiments are available at: <https://github.com/aderngram/heart-disease-borurank>.

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References

- World Health Organization. Cardiovascular diseases (CVDs) — Fact sheet. [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)), 2023.
- C. W. Tsao, A. W. Aday, Z. I. Almarzooq, et al. Heart disease and stroke statistics—2022 update: A report from the American Heart Association. *Circulation*, 145(8):e153–e639, 2022.
- A. Rajkomar, J. Dean, and I. Kohane. Machine learning in medicine. *New England Journal of Medicine*, 380(14):1347–1358, 2019.
- S. Mohan, C. Thirumalai, and G. Srivastava. Effective heart disease prediction using hybrid machine learning techniques. *IEEE Access*, 7:81542–81554, 2019.
- M. M. Ali, B. K. Paul, K. Ahmed, et al. Heart disease prediction using supervised machine learning algorithms: Performance analysis and comparison. *Computers in Biology and Medicine*, 136:104672, 2021.
- A. Dinh, S. Miertschin, A. Young, and S. D. Mohanty. A data-driven approach to predicting diabetes and cardiovascular disease with machine learning. *BMC Medical Informatics and Decision Making*, 19(1):1–15, 2019.
- A. Ullah, S. Uddin, and M. A. R. Siddiqui. A novel ensemble method for heart disease prediction using machine learning techniques. *Journal of Ambient Intelligence and Humanized Computing*, 2022.
- I. Guyon, J. Weston, S. Barnhill, and V. Vapnik. Gene selection for cancer classification using support vector machines. *Machine Learning*, 46(1–3):389–422, 2002.
- M. B. Kursu and W. R. Rudnicki. Feature selection with the Boruta package. *Journal of Statistical Software*, 36(11):1–13, 2010.

- M. T. Ribeiro, S. Singh, and C. Guestrin. “Why should I trust you?”: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, 2016.
- S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- A. B. Arrieta, N. Díaz-Rodríguez, J. Del Ser, et al. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58:82–115, 2020.
- L. Breiman. Bagging predictors. *Machine Learning*, 24(2):123–140, 1996.
- L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- P. Geurts, D. Ernst, and L. Wehenkel. Extremely randomized trees. *Machine Learning*, 63(1):3–42, 2006.
- R. Detrano, A. Janosi, W. Steinbrunn, et al. International application of a new probability algorithm for the diagnosis of coronary artery disease. *The American Journal of Cardiology*, 64(5):304–310, 1989.